

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (ME) Examination****November-December 2018****ME242/ME202.01/ME202 Engineering Thermodynamics****Date: 26.11.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.**

- (i) Classify following terms in category of Intensive and Extensive properties: **[02]**
 (a) Density (b) Specific Volume (c) Temperature (d) Viscosity
- (ii) Differentiate between Point function and Path function. **[02]**
- (iii) What is PMM-I? Is it possible to have PMM-I in actual practice? Justify your answer. **[02]**

Q – 2.a Define Exergy. Derive the equation of it for cyclic process with infinite heat source. [05]**Q – 2.b Answer any two questions. [10]**

- (i) Prove that “Internal Energy is a property of the system.”
- (ii) How the behavior of real gases deviates from ideal gas behavior? Explain Vander Wall’s equation of state for real gases.
- (iii) One kg of CO₂ has a volume of 1 m³ at 100°C. Compute the pressure by (i) Van der Waals’ equation (ii) Perfect gas equation. The values of a and b for CO₂:
 $a = 362850 \text{ Nm}^4/(\text{kg-mol})^2$ and $b = 0.0423 \text{ m}^3/\text{kg-mol}$, $R_0 = 8314 \text{ Nm/kg-mol K}$.

Q - 3 Answer any two questions. [14]

- (i) State general equation of Steady flow energy equation and explain each terms used in it. Derive the same for Diffuser, Steam Turbine and Condenser.
- (ii) A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exists from the nozzle. (b) If the inlet area is 0.1 m² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate. (c) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of the nozzle.
- (iii) A gas undergoes a thermodynamic cycle consisting of the following processes:

- (i) Process 1–2: Constant pressure $p_1 = 1.4$ bar, $V_1 = 0.028$ m³, $W_{12} = 10.5$ kJ
 (ii) Process 2–3: Compression with $pV = \text{constant}$, $U_3 = U_2$
 (iii) Process 3–1: Constant volume, $U_1 - U_3 = -26.4$ kJ. There are no significant changes in KE and PE. (a) Sketch the cycle on a p – V diagram (b) Calculate the net work for the cycle in kJ (c) Calculate the heat transfer for process 1–2. (d) Show that $\Sigma Q = \Sigma W$ for cycle.

SECTION – II

Q - 4 Answer the question below.

- (i) What is the highest possible theoretical efficiency of a heat engine operating with a hot reservoir of furnace gases at 2100°C when the cooling water available is at 15°C? [02]
 (ii) Draw T-S diagram of reversible and irreversible adiabatic compression process. [02]
 (iii) Define Coefficient of Volumetric expansion. Derive the equation of the same for ideal gas. [03]

Q – 5.a Derive any three Maxwell's Relation. [07]

OR

Q – 5.a With usual notations prove that, change in enthalpy of real gas, [07]

$$dh = C_p \times dT + \{v - T \times \beta \times v\} \times dp$$

Also derive the equation of change in enthalpy for ideal gas from above equation.

Q – 5.b Prove that Clausius and Kelvin Plank Statements are equivalent to each other. [06]

Q – 5.c An inventor claims to have developed an engine that takes in 105 MJ at a temperature of 400 K, rejects 42 MJ at a temperature of 200 K, and delivers 15 kWh of mechanical work. Would you advise investing money to put this engine in the market? [05]

OR

Q – 5.c Two reversible heat engines A and B are arranged in series, A rejecting heat directly to B. Engine A receives 200 kJ at a temperature of 421°C from a hot source, while engine B is in communication with a cold sink at a temperature of 4.4°C. If the work output of A is twice that of B, find (a) The intermediate temperature between A and B (b) The efficiency of each engine (c) The heat rejected to the cold sink. [05]

Q – 6.a State and prove Clausius Equality. [05]

Q – 6.b A block of iron weighing 100 kg and having a temperature of 100°C is immersed in 50 kg of water at a temperature of 20°C. What will be the change of entropy of the [05]

combined system of iron and water? Specific heats of iron and water are 0.45 and 4.18 kJ/kg K respectively.

OR

Q – 6.a Prove the statement “Entropy of Universe always increases.” **[05]**

Q – 6.b Define Entropy. Derive the equation of change in entropy for open system. **[05]**
